Title: 输入信号 → LTI → 输出信号　　卷积的定义与图解法　　Date: / /

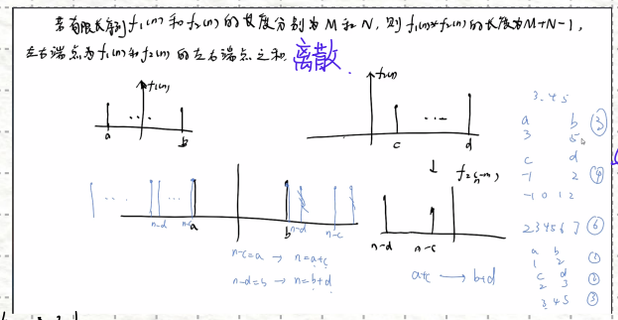
zero state

卷积 $f_1(t)*f_2(t)\triangleq\int_{-\infty}^{+\infty}f_1(\tau)f_2(t-\tau)d\tau=y_{zs}(t)$

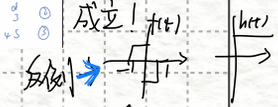
$\delta(t)\to$ LTI $\to h(t)$ 冲激响应 只与sys特性有关

$u(t)\to$ LTI $\to g(t)/s(t)$ 阶跃响应

$\begin{cases}\frac{dg(t)}{dt}=h(t)\Rightarrow\int_{-\infty}^{t}h(\tau)d\tau=g(t)\\ \frac{du(t)}{dt}=\delta(t)\Rightarrow\int_{-\infty}^{t}\delta(\tau)d\tau=u(t)\end{cases}$



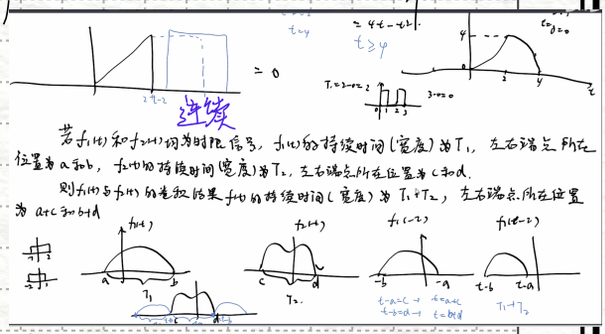
对非因果不一定成立！

反例 

$f(t)\to y_{zs}(t)$　　时线性系统 → $f(t-\Delta t)\to y_{zs}(t-\Delta t)$

推 $f'(t)\to y'_{zs}(t)$. ← $\lim\frac{f(t)-f(t-\Delta t)}{\Delta t}\to\lim\frac{y_{zs}(t)-y_{zs}(t-\Delta t)}{\Delta t}$

积分性: $\int_{-\infty}^{t}f(\tau)d\tau\to\int_{-\infty}^{t}y_{zs}(\tau)d\tau$



e.g. $y(t)=\int_{-\infty}^{+\infty}f(\tau)h(t-\tau)d\tau=f(t)*h(t)$ (连续卷积)

$t\,u(t)*u(t)=?$　　由 $u(t)=\begin{cases}1 & t>0\\ 0 & t<0\end{cases}$

$\Rightarrow\int_{-\infty}^{+\infty}\tau\,u(\tau)\cdot u(t-\tau)d\tau$　　$u(t-\tau)=\begin{cases}1 & t-\tau>0\ \text{即}\ \tau<t\\ 0 & t-\tau<0\ \text{即}\ \tau>t\end{cases}$

$\Rightarrow u(\tau)\cdot u(t-\tau)=\begin{cases}1 & 0<\tau<t\\ 0 & \tau<0\end{cases}$

$\Rightarrow y(t)=\int_0^t\tau\,d\tau=\frac{1}{2}\tau^2\Big|_0^t=\frac{1}{2}t^2$ (当$t<0$时，$0<\tau<t$不成立即在积分结果后乘上$u(t)$来解决)

$\Rightarrow y(t)=\frac{1}{2}t^2u(t)$

(亦可由图解法得)
1. 简单求面积
2. 函数式相乘求积分

[离散卷积] 与之类似，结果乘上 $u[n]$

$x[n]\to$ h[n] $\to y[n]$

$y[n]=\sum_{k=-\infty}^{+\infty}x[k]\cdot h[n-k]=x[n]*h[n]$ 卷积和/叠加和

$y(t)=\int_{-\infty}^{+\infty}x(\tau)h(t-\tau)d\tau=x(t)*h(t)$ 卷积积分/叠加积分

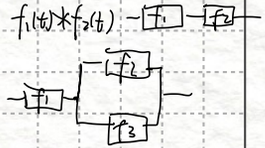
交换律: $x(t)*h_1(t)=h_1(t)*x(t)$

分配律 $f_1(t)*[f_2(t)+f_3(t)]=f_1(t)*f_2(t)+f_1(t)*f_3(t)$

$f_1(t)*Af_2(t)=A[f_1(t)*f_2(t)]$

结合律 $f_1(t)*f_2(t)*f_3(t)=f_1(t)*[f_2(t)*f_3(t)]$ (既有乘积也有卷积，不能用结合律)

两sys相互卷积 ⇒ 串联
两sys相加 ⇒ 并联



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